

Sandeep Singh R

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Professional Summary

Strategic and results-driven Project & Product Manager with over 10 years of experience in Embedded systems, IoT, and product lifecycle management. Adept at executing complex technical projects, optimizing product roadmaps, and driving operational efficiency. Skilled in Agile methodologies, PDLC, and embedded product delivery, with proven expertise in stakeholder management, budget optimization, cross-functional collaboration, and regulatory compliance (ISO, DO-178). Completed Product Management Certification from XLRI and Certified PMP to further strengthen project management credentials

Core Skills

- **Project & Product Management:** Agile & Scrum, Program Management, PDLC, Stakeholder Management, Risk Assessment, Budgeting & Cost Control, Vendor Negotiation, Resource Allocation, P&L Management
- **Technical Expertise:** Embedded Systems, IoT, C, Embedded C, Python, MATLAB, Firmware Development
- **Process & Quality:** Workflow Automation, Quality Assurance, Technical Documentation, Design Reviews, Compliance (ISO, DO-178, MISRA C)
- **Leadership & Tools:** Team Building, Cross-Functional Collaboration, JIRA, Bitbucket, Git, PCB Design (KiCad, Altium)
- **Domain :** Semiconductor, Automation

Professional Experience

Project Manager

Klug Avalon Mechatronics Pvt. Ltd. | Jun 2023 – Present

- **Reduced crane downtime incidents by 40%** by leading an 8-member engineering team in the design and deployment of **Safe Load Indicator (SLI) systems**.
- **Achieved 100% on-time delivery** of 5+ major projects through effective program management, resource allocation, and Agile execution.
- **Increased client satisfaction to 95%** by aligning stakeholder expectations with project goals and conducting structured review meetings.
- Optimized cross-functional collaboration, improving engineering workflow efficiency by **30%**.

Assistant Manager – R&D

MPA Pvt. Ltd. | May 2022 – Jun 2023

- **Cut development cycles by 20%** by implementing Agile practices and embedded system optimization.
- **Boosted device performance by 25%** through firmware enhancements and efficient resource utilization.
- Maintained **95% client satisfaction** by driving collaboration between product and engineering teams.
- Improved vendor coordination processes, enhancing **component sourcing efficiency by 15%**.

Senior Embedded Design Engineer

STPL | Mar 2021 – May 2022

- **Improved system stability by 30%** through effective integration of hardware and software modules.
- **Reduced defects by 40%** by introducing enhanced QA processes, design reviews, and structured validation.
- Supported execution of the **product roadmap**, aligning embedded features with customer requirements.

Senior Software Developer (Client: Microchip Pvt. Ltd.)

Nexsemi Systems Pvt. Ltd. | Oct 2018 – Nov 2019

Microchip Projects:

- **Validation Engineer (Mar 2019 – Sep 2019)**
 - **Improved testing efficiency by 80%** by automating manual test cases with Jython scripting.
 - **Enhanced bug detection by 25%** through structured test planning and improved coverage.
 - Developed reusable validation frameworks, reducing regression testing efforts by 15%.
- **MATLAB Developer (Oct 2018 – Mar 2019)**
 - **Increased visualization efficiency by 35%** by developing advanced MATLAB-based GUI tools.
 - Prototyped algorithms for embedded systems, enabling **faster simulation and performance testing**.
 - Delivered MATLAB dashboards that accelerated engineering decision-making.

Core Contributions (Nexsemi Role):

- **Enhanced product efficiency by 20%** by delivering critical embedded software updates.
- **Reduced deployment errors by 25%** by streamlining version control workflows and automating CI/CD processes.
- Collaborated with cross-functional teams to accelerate release cycles and improve reliability.

Embedded Design Engineer

Monee & Company Pvt. Ltd. | Nov 2016 – Nov 2017

- **Improved system stability by 20%** by designing and testing embedded hardware solutions.
- Designed PCB layouts and firmware modules that reduced prototyping timelines by **15%**.
- Partnered with clients to customize embedded solutions, improving customer adoption rates.

Research Associate

PESIT | Oct 2012 – Aug 2014

- **Enhanced data accuracy by 95%** through research on acoustic systems and algorithm development.
- Published findings in **5 national conferences and 2 international journals**, contributing to academic recognition.
- Mentored undergraduate teams, supporting research and embedded systems development.

Design Engineer – Embedded

Apsis Solutions | Jun 2011 – Oct 2012

- **Improved embedded performance by 15%** by designing optimized hardware solutions.
- **Reduced hardware-software integration issues by 20%** through systematic debugging and testing.
- Created technical documentation and feasibility studies that supported client project proposals

Key Projects

Safe Load Indicator System for Cranes

- - Reduced crane safety incidents by 40%.
- - Enhanced calibration processes, cutting setup time by 25%.

Spectrometer Design and Development

- - Developed STM32 firmware, improving data acquisition accuracy by 30%.
- - Optimized testing methodologies, reducing hardware integration time by 20%.

Automatic Flight Control System (AFCS)

- - Implemented FreeRTOS integration that improved system responsiveness by 15%.
- - Led hardware rig testing, reducing system failures by 25%.

Education & Certifications

Degree

- M.Tech (Signal Processing) | SJCT, VTU | 2016
- B.E (Electronics & Communication) | SSEC, VTU | 2011

Certification

- Product Management Certification – XLRI
- PMP®
- Embedded Systems and Peripherals – Microchip India Pvt. Ltd.
- Online Certifications – Project Management & Embedded Systems (Coursera)

Achievements & Extracurricular Activities

- Winner – Best Design in Embedded Design Contest (EFY)
- Published 5 National Conference Papers & 2 International Journals
- MATLAB Instructor – Udemy (Basic Course)
- NCC 'B' Certificate holder
- Conducted multiple workshops and training sessions for engineering students
- Passionate about adventure sports: rock climbing, rappelling, and trekking

Patents

- **Automobile Emergency Alerting Autonomous Device** – *Design Patent Filed (Registration No. 453281-001, India), Mar 2025*

Additional Information

DOB: 4th April 1986

Passport: K2691395

PAN: DNLPS6372G

Location: Bengaluru, Karnataka, India (Currently in Pune, Maharashtra)